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# Memory-Gravity Attention: A Persistent Attention Mechanism for Long-Horizon Agent Memory

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[github.com/jkdkr2439/Memory-Gravity-Attention](https://github.com/jkdkr2439/Memory-Gravity-Attention)

## Abstract

Standard attention mechanisms allocate weights over tokens within a bounded context window, optimizing for local relevance. This is insufficient for long-horizon agents that must decide which prior information remains structurally important across time. We propose Memory-Gravity Attention (Memory-Gravity Attention), a persistent attention mechanism over dynamic memory graphs. Each memory node maintains a gravity score combining semantic similarity with accumulated structural importance: historical weight, recency, frequency, unresolvedness, goal relevance, utility, and error penalties. We introduce attention entropy to detect two failure modes—attention lock (repeated narrow retrieval) and attention diffusion (unfocused retrieval)—and propose control mechanisms for each.

We validate MGA on a controlled benchmark (10 synthetic worlds, 4,500+ nodes, 415 queries, sentence-transformer embeddings). The Generative-Agents baseline [Park et al., 2023] achieves the highest overall Recall@5 (0.209) but scores **0.000** on needle, open-loop, and noise-resistance query families—precisely the tasks requiring persistent memory. MGA achieves non-zero recall on all three (needle: 0.143, open-loop: 0.222, noise: 0.075) with the lowest noise rate (0.017). Learned coefficients confirm goal-relevance as the dominant persistent signal ( $\theta = 2.92$ ). Entropy diagnostics confirm no pathological lock behavior. Code: <https://github.com/jkdkr2439/Memory-Gravity-Attention>.

## 1 Introduction

Transformer attention answers: “Which representations in the current context are relevant to my query?” This is powerful for sequence modeling but insufficient for long-horizon agents. Such agents face a different question: “Which memory nodes have accumulated enough importance to influence the current state, regardless of recency or surface similarity?”

Consider failure modes of current memory retrieval:

- A fact stated once in session 2 of 25, never repeated, but still needed (**needle**).
- A task opened early and never closed (**open-loop**).
- A constraint revised mid-project where the old version has high similarity (**stale trap**).
- Dense logs with 40%+ chatter (**noise resistance**).

Existing systems use context-window attention, vector similarity, recency, summarization, or static profiles. These are useful but incomplete. Semantic similarity alone cannot determine long-term importance. Recency alone cannot identify structurally important information. The missing mechanism is *persistent salience*: a way to assign and update long-term importance over memory nodes.

Park et al. [Park et al., 2023] proposed  $\alpha \cdot \text{recency} + \beta \cdot \text{importance} + \gamma \cdot \text{relevance}$  for generative agents. As we show empirically, this still scores zero on needle, open-loop, and noise-resistance tasks.

We propose Memory-Gravity Attention: a persistent attention mechanism that scores memory by combining local relevance with accumulated structural importance, and introduces entropy-based diagnostics to detect and correct pathological retrieval patterns.

## 2 Related Work

**Memory for LLM agents.** Park et al. [Park et al., 2023] use a three-term additive scoring. MemGPT [Packer et al., 2023] manages memory via virtual context. Reflexion [Shinn et al., 2023] uses verbal self-reflection. None benchmark persistent-importance retrieval.

**RAG.** Standard RAG [Lewis et al., 2020] uses dense similarity. Re2G [Glass et al., 2022] adds reranking. These focus on factual QA, not persistent memory management.

**Novelty-driven attention.** Free-energy [Friston, 2010] and Bayesian surprise [Itti and Baldi, 2009] attend to prediction errors. Differential Transformer [Ye et al., 2024] reduces noise via attention subtraction. These modify transformer internals; MGA operates as an external retrieval layer.

## 3 Memory-Gravity Attention: Theory

### 3.1 Agent and Memory Graph

Let  $\mathcal{A}$  denote a memory-augmented agent. At time  $t$ , it receives input  $x_t$ , maintains state  $S_t$ , retrieves memory  $R_t$ , forms context  $C_t$ , generates output  $y_t$ , and updates state:

$$x_t \rightarrow S_t \rightarrow R_t \rightarrow C_t \rightarrow y_t \rightarrow S_{t+1}.$$

The agent maintains a memory graph  $\mathcal{M}_t = (\mathcal{N}_t, \mathcal{E}_t)$  where each node  $n_i$  stores content, embedding, and persistent metadata.

### 3.2 Memory Node

Each node carries:

$$n_i = (c_i, e_i, w_i, s_i, r_i, f_i, u_i, g_i, p_i, q_i, \epsilon_i, \tau_i, k_i)$$

where  $c_i$  is content,  $e_i$  is embedding,  $w_i$  is persistent weight,  $s_i$  is salience,  $r_i$  is recency,  $f_i$  is frequency,  $u_i$  is unresolvedness,  $g_i$  is goal relevance,  $p_i$  is profile relevance,  $q_i$  is utility,  $\epsilon_i$  is error penalty,  $\tau_i$  is timestamp, and  $k_i$  is cluster ID.

### 3.3 Gravity Score

For input encoded as  $z_t = E(x_t)$ , the gravity score for node  $n_i$  is:

$$\begin{aligned} G_i(t) = & \lambda_s \text{Sim}(z_t, e_i) + \lambda_w W_i(t) + \lambda_r R_i(t) + \lambda_f F_i(t) \\ & + \lambda_u U_i(t) + \lambda_g \text{Goal}_i(t) + \lambda_p \text{Profile}_i(t) \\ & + \lambda_q Q_i(t) + \lambda_a \text{Salience}_i(t) - \lambda_e \text{Error}_i(t) \end{aligned} \quad (1)$$

The attention distribution over memory:

$$A_i(t) = \frac{\exp(G_i(t))}{\sum_j \exp(G_j(t))} = \text{softmax}(G(t))_i \quad (2)$$

### 3.4 Component Definitions

- **Similarity:**  $\text{Sim}(z_t, e_i) = \cos(z_t, e_i)$
- **Recency:**  $R_i(t) = \exp(-(t - t_i^{\text{last}})/\tau_r)$
- **Frequency:**  $F_i(t) = \log(1 + \text{count}_i(t))$

- **Unresolvedness:**  $U_i(t) \in [0, 1]$ , detecting open loops
- **Goal relevance:**  $\text{Goal}_i(t) = \cos(g_t, e_i)$  where  $g_t$  encodes the active goal
- **Utility:**  $Q_i(t) = \text{ExpectedUseful}(n_i \mid \text{task}_t)$ , from feedback
- **Error penalty:**  $\text{Error}_i(t) \geq 0$ , from stale/contradictory/irrelevant retrieval
- **Historical weight:**  $W_i(t)$ , accumulated structural importance

### 3.5 Memory Weight Update

After output and feedback:

$$W_i(t+1) = \rho W_i(t) + \alpha A_i(t) Q_i(t) + \beta A_i(t) \text{Saliency}_i(t) + \delta A_i(t) U_i(t) + \eta A_i(t) \text{Goal}_i(t) - \gamma \text{Error}_i(t) \quad (3)$$

where  $\rho \in (0, 1)$  is decay,  $\alpha, \beta, \delta, \eta$  are reinforcement rates, and  $\gamma$  is error penalty strength.

### 3.6 Cluster-Level Gravity

Nodes group into clusters  $\mathcal{C} = \{C_1, \dots, C_m\}$ . Cluster gravity:

$$G_{C_j}(t) = \sum_{n_i \in C_j} A_i(t)$$

Cluster repetition rate over window  $L$ :

$$\text{CRR}(C_j, t, L) = \frac{1}{L} \sum_{\ell=t-L+1}^t \mathbb{I}[C_j = \arg \max_{C_k} G_{C_k}(\ell)]$$

## 4 Attention Entropy and Control

### 4.1 Entropy Definition

$$\mathcal{H}(A_t) = - \sum_{i=1}^{|\mathcal{N}_t|} A_i(t) \log A_i(t) \quad (4)$$

### 4.2 Attention Lock

Detected when entropy is consistently low:

$$\mathcal{H}(A_t) < \theta_{\text{low}} \quad \text{for } k \text{ consecutive steps}$$

Combined with cluster dominance:  $\text{CRR}(C_j, t, L) > \theta_{\text{cluster}}$ .

Indicates: repeated reasoning loops, fixation on one hypothesis, over-personalization.

**Response:** suppress dominant cluster ( $G'_i(t) = G_i(t) - \mu \cdot \mathbb{I}[n_i \in C_{\text{locked}}]$ ), retrieve from alternatives, switch to explore mode.

### 4.3 Attention Diffusion

Detected when entropy is consistently high:  $\mathcal{H}(A_t) > \theta_{\text{high}}$  for  $k$  steps.

Indicates: unclear objective, noisy memory, weak specificity.

**Response:** increase  $\lambda_g$ , reduce top- $k$ , prune stale nodes, prioritize unresolved tasks.

### 4.4 Target Operating Zone

$$\theta_{\text{low}} < \mathcal{H}(A_t) < \theta_{\text{high}} \quad (5)$$

The goal is adaptive focus—neither locked nor diffuse.

## 5 MGA Gate: A Practical Instantiation

For empirical validation, we implement a simplified variant—**MGA Gate**—that preserves the core theoretical property (similarity gates retrieval, persistent importance amplifies):

$$\text{score}_i = \text{sim}_i \times (1 + \sigma(\boldsymbol{\theta}^\top \mathbf{f}_i)) \quad (6)$$

where  $\mathbf{f}_i = [\text{recency}, \text{freq}, \text{unresolved}, \text{utility}, \text{weight}, \text{goal\_rel}] \in [0, 1]^6$  and  $\boldsymbol{\theta} \in \mathbb{R}^6$  is learned via logistic regression on training queries.

**Key property:** The multiplicative gate ensures  $\text{score}_i \in [\text{sim}_i, 2 \cdot \text{sim}_i]$ . Low-similarity nodes *cannot* be dragged in by importance alone. Among nodes with comparable similarity, persistent importance disambiguates.

**Relation to full theory:** Eq. 6 corresponds to the full gravity score (Eq. 1) with: (a) similarity factored out multiplicatively rather than additively, (b) profile relevance, salience, and error penalty set to zero (uninformative in benchmark), and (c) coefficients learned from data rather than hand-set.

**Ablation:** We also test **MGA Linear** ( $\boldsymbol{\theta}^\top [\text{sim}, \mathbf{f}]$ ) to isolate the gate structure’s contribution.

## 6 Experimental Setup

### 6.1 Synthetic Benchmark

10 synthetic worlds, each with: 25 sessions; 10–25 events/session ( $\sim 450$  nodes); 3 domains (research, software, design); 2–4 mid-log revisions (stale traps); 4–8 tasks with staggered open/close; 45% chatter (noise); 5–8 templates per event type.

Query families: constraint recall, stale trap, open-loop, needle, noise resistance. Train/test: 40%/60%.  $\boldsymbol{\theta}$  learned on train only.

### 6.2 Embeddings and Evaluation

Sentence-transformers (all-MiniLM-L6-v2), GPU (NVIDIA T4). Metrics: Recall@5, nDCG@5, Stale@5 (fraction of retrieved that are superseded), Noise@5 (fraction that are chatter).

### 6.3 Baselines

- **Recency:** rank by  $R_i(t)$  only.
- **Similarity:** rank by  $\text{Sim}(z_t, e_i)$  only.
- **GA** [Park et al., 2023]:  $\text{sim} + \text{recency} + \text{weight}$  (additive).
- **MGA Gate:** Eq. 6 (multiplicative).
- **MGA Linear:**  $\boldsymbol{\theta}^\top [\text{sim}, \mathbf{f}]$  (ablation).

## 7 Results

### 7.1 Overall Performance

Table 1: Retrieval performance (415 test queries, 10 worlds). Best in **bold**.

| Retriever  | Recall@5     | nDCG@5       | Stale@5      | Noise@5      |
|------------|--------------|--------------|--------------|--------------|
| Recency    | 0.029        | 0.085        | 0.000        | 0.364        |
| Similarity | 0.138        | 0.496        | 0.340        | 0.046        |
| GA         | <b>0.209</b> | <b>0.697</b> | <b>0.000</b> | 0.070        |
| MGA gate   | 0.174        | 0.562        | 0.265        | 0.029        |
| MGA linear | 0.195        | 0.614        | 0.205        | <b>0.017</b> |

## 7.2 Per-Family Breakdown

Table 2: Recall@5 by query family. **Bold** = best per row. GA scores 0.000 on all persistent-memory families.

| Family       | Recency | Sim   | GA           | MGA gate     | MGA linear   |
|--------------|---------|-------|--------------|--------------|--------------|
| Constraint   | 0.036   | 0.151 | <b>0.257</b> | 0.191        | 0.220        |
| Needle       | 0.000   | 0.143 | 0.000        | <b>0.143</b> | 0.000        |
| Open-loop    | 0.000   | 0.097 | 0.000        | 0.111        | <b>0.222</b> |
| Noise resist | 0.000   | 0.050 | 0.000        | <b>0.075</b> | 0.025        |
| Stale trap   | 0.036   | 0.135 | <b>0.275</b> | 0.187        | 0.230        |

## 7.3 Learned Feature Importance

Table 3: Learned  $\theta$  (mean  $\pm$  std, 10 worlds). Goal relevance dominates.

| Feature        | Coefficient      |
|----------------|------------------|
| Goal relevance | $2.92 \pm 0.87$  |
| Recency        | $0.67 \pm 0.50$  |
| Frequency      | $0.40 \pm 0.35$  |
| Unresolvedness | $0.21 \pm 0.29$  |
| Utility        | $-0.01 \pm 0.01$ |
| Weight         | $-0.01 \pm 0.01$ |

This empirically confirms the theoretical prediction: knowing what the system is currently working on ( $\lambda_g$ ) is the most important persistent signal.

## 7.4 Attention-Entropy Diagnostic

We compute normalized entropy  $\hat{H} = \mathcal{H}(A_t)/\log N$  and top-5 concentration:

Table 4: Entropy diagnostic (5 worlds, 29 queries/retriever). No pathological lock detected.

| Retriever  | Mean $\hat{H}$ | Std $\hat{H}$ | Top-5 Conc. |
|------------|----------------|---------------|-------------|
| Recency    | 0.992          | 0.001         | 0.050       |
| Similarity | 0.992          | 0.005         | 0.060       |
| GA         | 0.983          | 0.005         | 0.078       |
| MGA gate   | 0.976          | 0.014         | 0.093       |
| MGA linear | 0.650          | 0.131         | 0.467       |

All retrievers operate within the healthy zone ( $\hat{H} > \theta_{\text{low}}$ ). MGA linear concentrates most aggressively (0.467 top-5 concentration), consistent with its role as a direct importance-weighted scorer. MGA gate maintains high entropy (0.976) while still achieving selective boosting—the multiplicative structure amplifies without collapsing attention.

# 8 Discussion

## 8.1 Theory Predicts, Experiment Confirms

- **Theory:** Goal relevance ( $\lambda_g$ ) should be a key persistent signal. **Experiment:**  $\theta_{\text{goal}} = 2.92$ , dominant coefficient.
- **Theory:** Additive scoring (GA) lacks structural separation between similarity and importance. **Experiment:** GA scores 0.000 on persistent-memory families.

- **Theory:** Multiplicative gating prevents low-similarity contamination. **Experiment:** MGA gate achieves non-zero recall on all families while maintaining low noise.
- **Theory:** Attention entropy detects lock. **Experiment:** No retriever exhibits pathological lock; MGA gate stays in healthy zone.

## 8.2 Why GA Fails on Persistent Tasks

GA combines signals additively: a node with high recency can dominate even with low relevance. On needle queries (old facts), recency suppresses. On open-loop queries, the importance proxy is uninformative without task-tracking. On noise queries, no mechanism distinguishes signal from chatter beyond similarity.

MGA’s gate structure ensures similarity is necessary for retrieval (gating), while persistent features amplify among similarly-relevant candidates (boosting).

## 8.3 Limitations

1. **Synthetic benchmark:** template-based logs lack real-world lexical diversity.
2. **Simplified instantiation:** benchmark tests 6 of 10 theoretical factors (profile, salience, error penalty uninformative in current setup).
3. **Small training signal:** 4–5 queries/world for  $\theta$  estimation.
4. **GA wins overall:** MGA’s advantage is specifically on long-horizon tasks.
5. **No online weight update:** current benchmark uses static features; the full update rule (Eq. 5) requires multi-session deployment.

## 8.4 Broader Significance

MGA operationalizes a structural principle: an agent should attend not only to what is *similar* to its current query, but to what is *persistently important* for ongoing operation. This parallels gap-based attention in biological systems—survival-relevant information receives priority regardless of surface similarity.

## 9 Conclusion

We introduced Memory-Gravity Attention, a persistent attention mechanism with both a complete theoretical framework and empirical validation:

1. **Theory:** gravity scoring over 10 factors, weight update dynamics, cluster-level analysis, entropy diagnostics, and control mechanisms.
2. **Experiment:** Standard approaches score zero on persistent-memory tasks. MGA achieves non-zero recall on all such tasks.
3. **Insight:** Goal-relevance ( $\theta = 2.92$ ) is the dominant signal. The multiplicative gate maintains attention diversity while enabling selective boosting.
4. **Diagnostic:** Entropy monitoring confirms healthy behavior and provides a principled tool for detecting failure modes in deployed systems.

MGA does not replace similarity-based retrieval. It fills the structural gap where similarity and recency fail: persistent, long-horizon, noise-resistant memory.

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